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Astorga et al.

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(54) **ANTENNA ARRAY ARCHITECTURE WITH ELECTRICALLY CONDUCTIVE COLUMNS BETWEEN SUBSTRATES**

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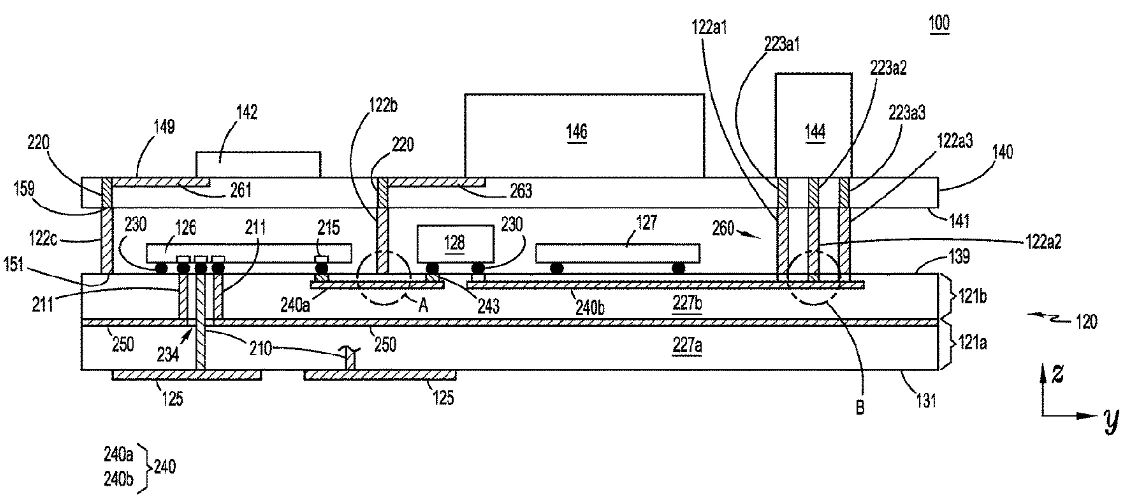
(57) **ABSTRACT**

An antenna apparatus includes an antenna substrate with opposite first and second surfaces; and a PCB having opposite first and second surfaces. Antenna elements are disposed at the first surface of the antenna substrate. Electrically conductive columns, each having a first end attached to the second surface of the PCB and a second end attached to the second surface of the antenna substrate, secure the PCB to the antenna substrate and provide an electrical interconnect between the PCB and the antenna substrate. RFIC chips are each attached to the second surface of the antenna substrate and are coupled to the antenna elements. At least one circuit element is attached to the first surface of the PCB and electrically coupled to at least one of the RFIC chips through at least one of the electrically conductive columns.

24 Claims, 13 Drawing Sheets



(51) **Int. Cl.**

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H01R 12/52	(2011.01)
H05K 1/14	(2006.01)
H05K 1/18	(2006.01)

(52) **U.S. Cl.**

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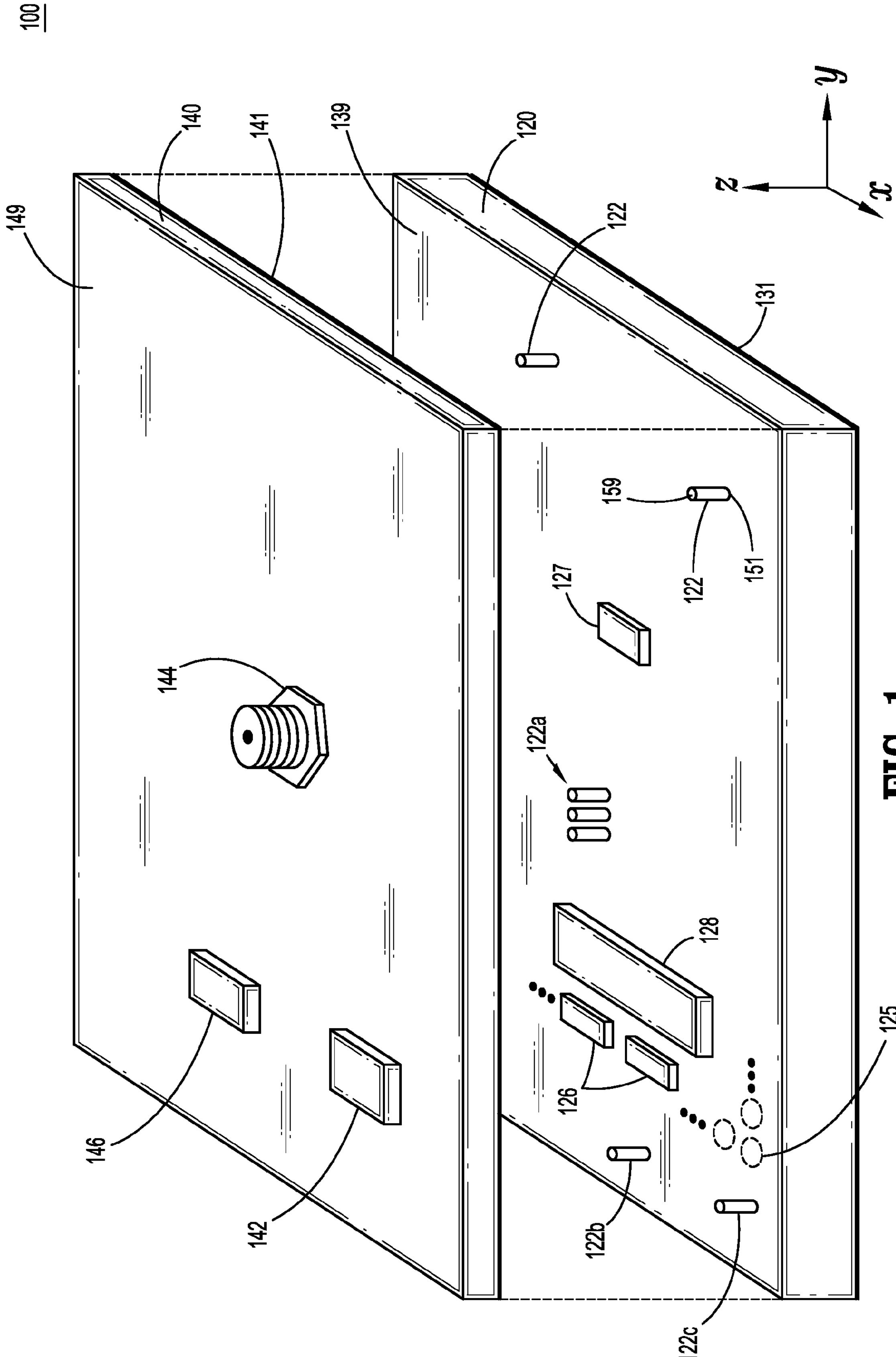
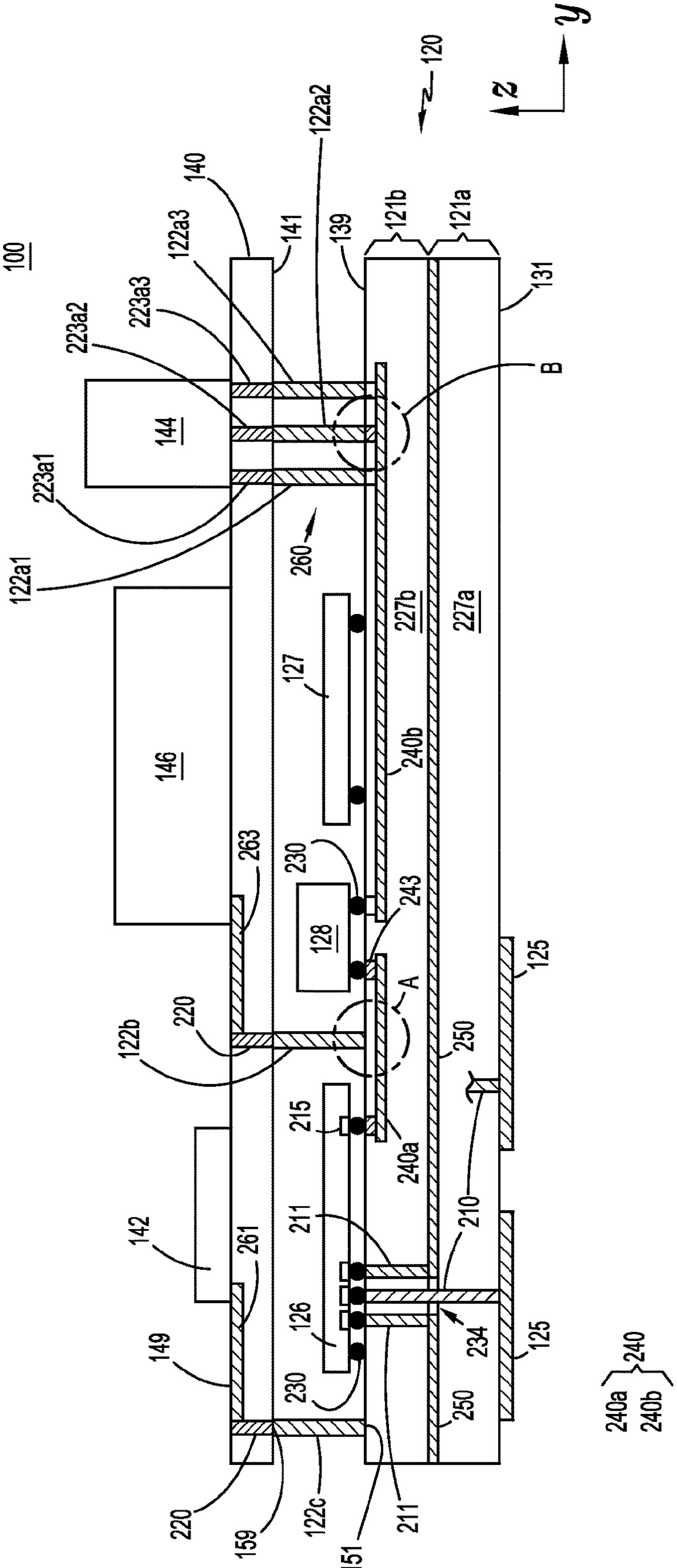

100
149
140
141
139
120
122
131
159
122
151
127
144
122a
128
146
142
126
125
122b
122c
y
z
x

FIG. 1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120
y
z
140
141
139
121b
121a
131
122a2
122a3
223a3
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144
B
260
227b
227a
127
146
240b
263
230
243
128
125
A
220
122b
250
215
240a
210
211
142
261
126
234
149
230
250
125
240
240a
240b
220
159
122c
151
211

FIG. 2

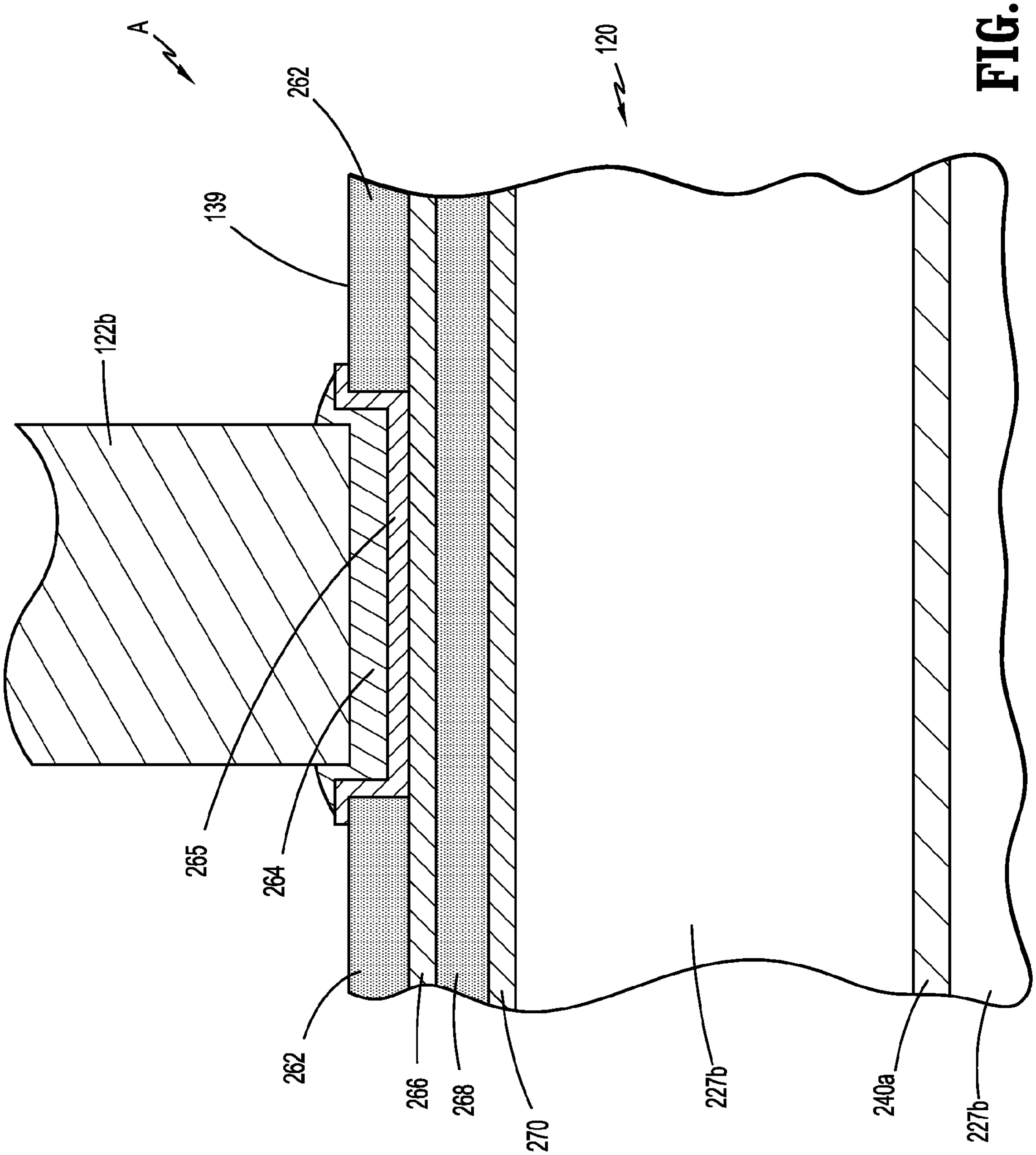


FIG. 3A

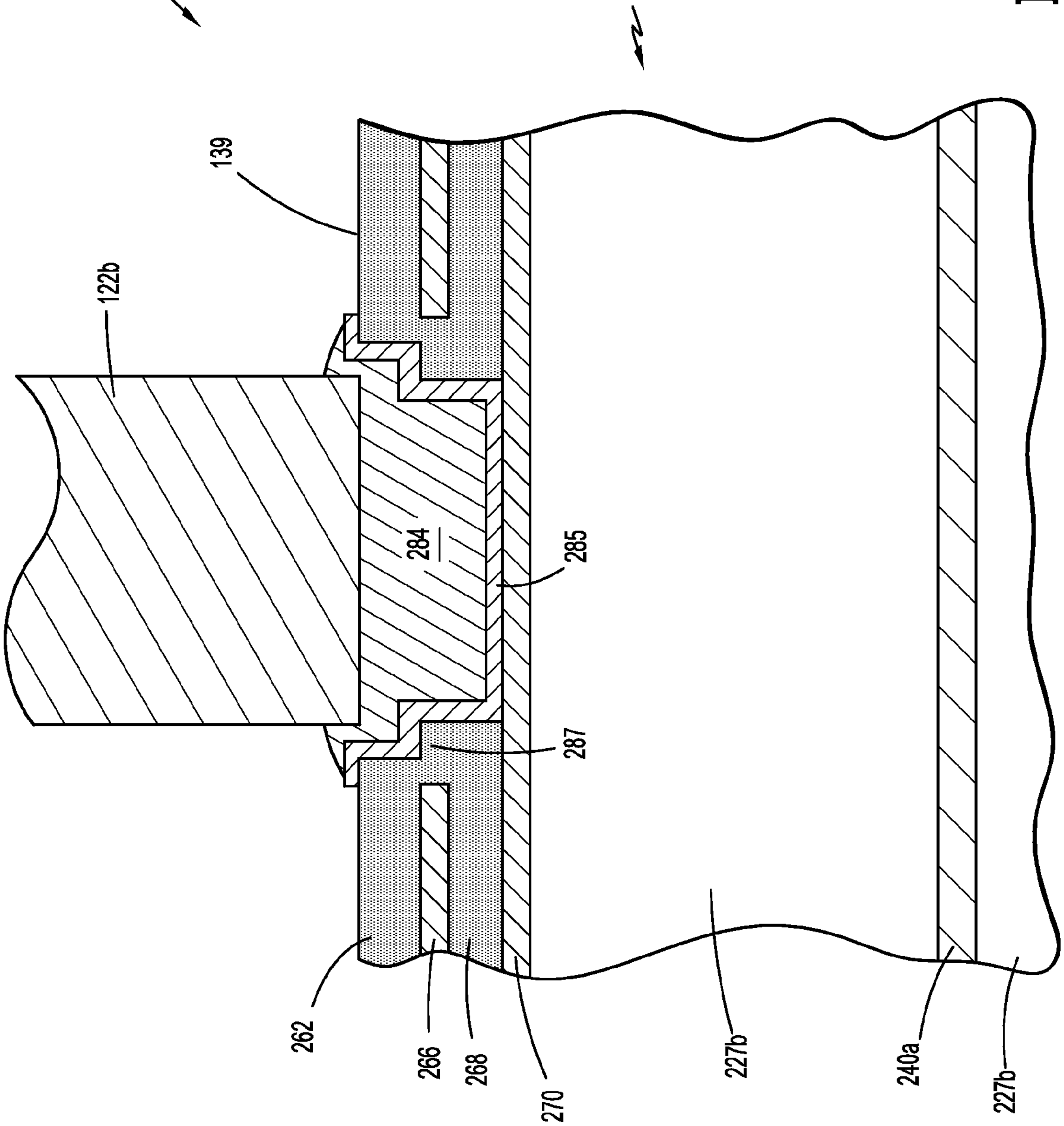


FIG. 3B

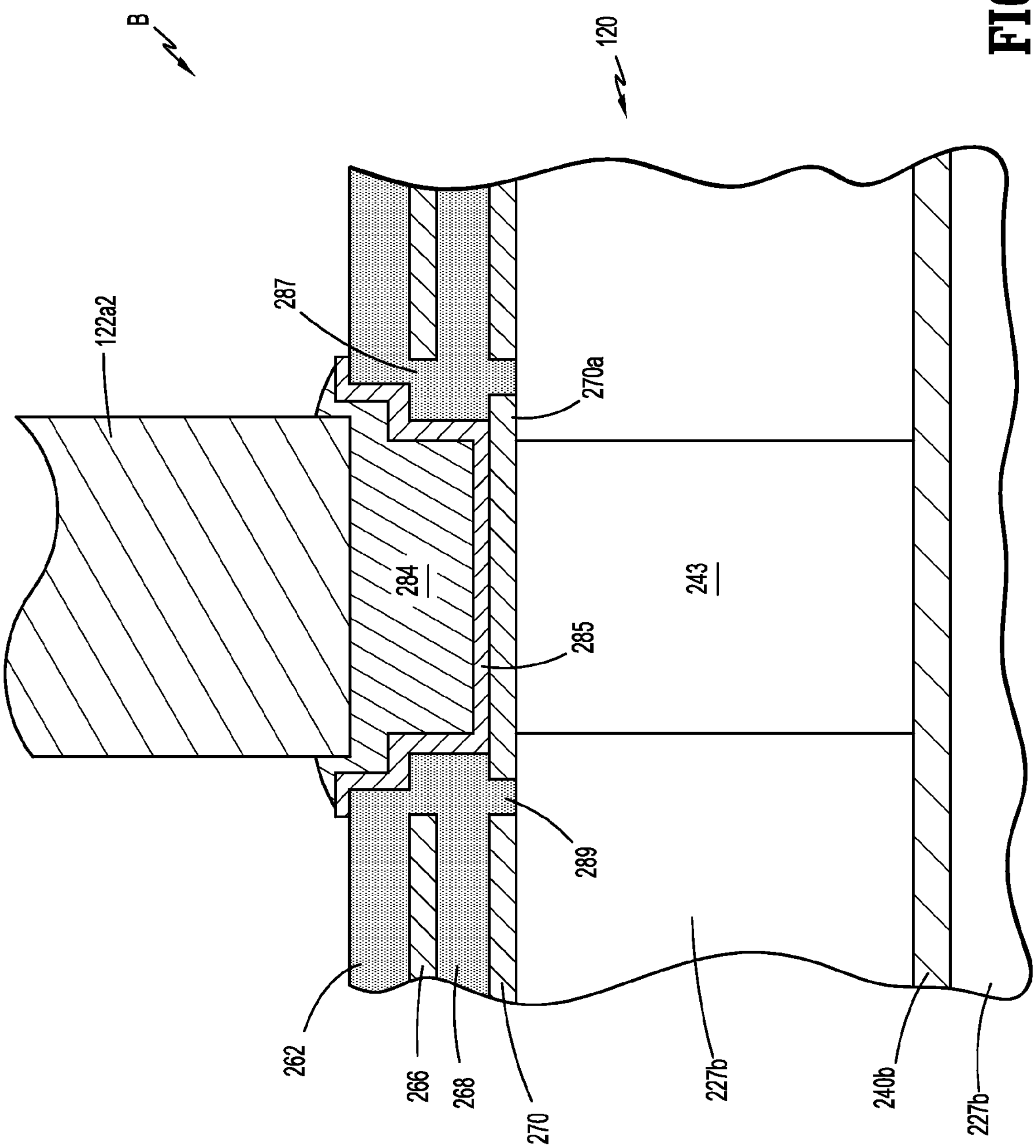
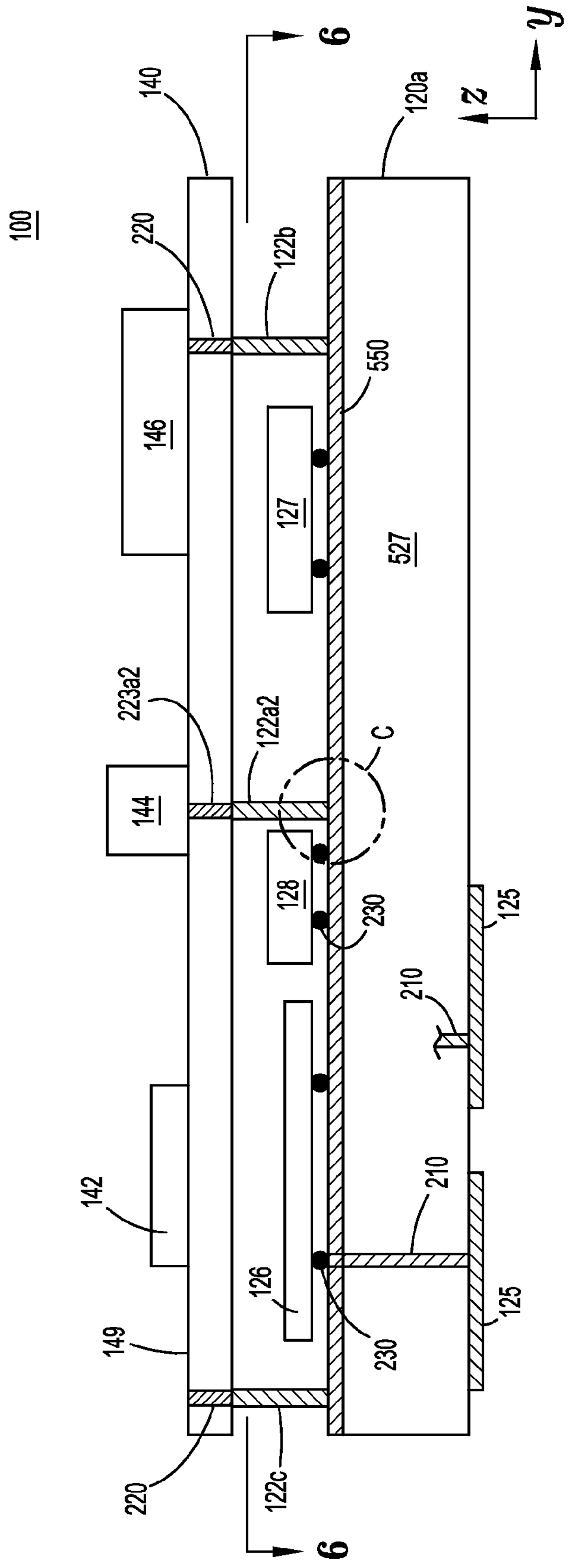

B
120
122a2
287
270a
284
243
285
289
262
266
268
270
227b
240b
227b

FIG. 4



100
140
6
z
y
120a
220
122b
146
550
127
527
223a2
122a2
C
144
128
230
125
210
142
126
149
230
220
122c
6

FIG. 5A

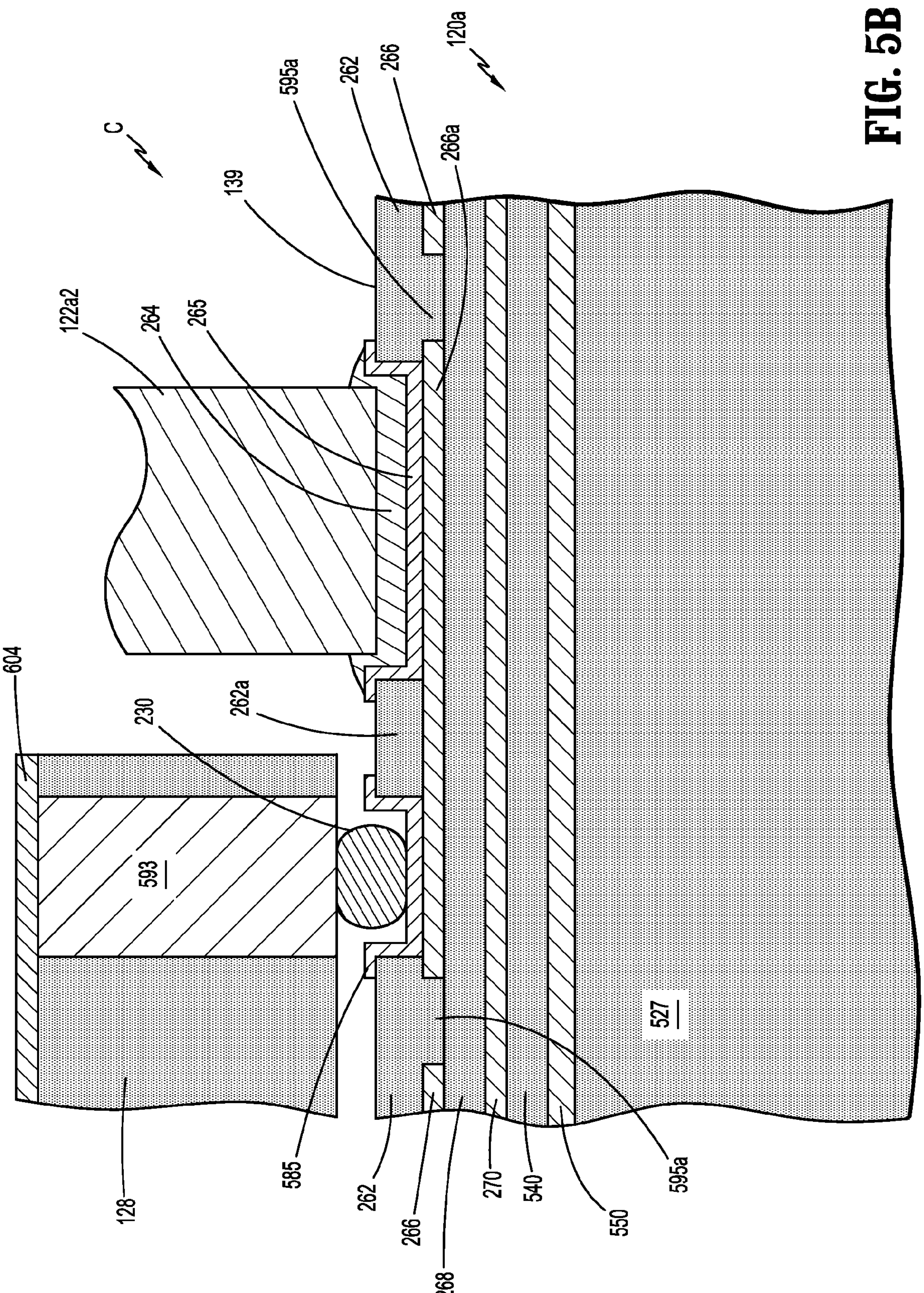

C
120a
595a
262
266
266a
139
122a2
264
265
604
230
262a
593
527
128
585
262
266
268
270
540
550
595a

FIG. 5B

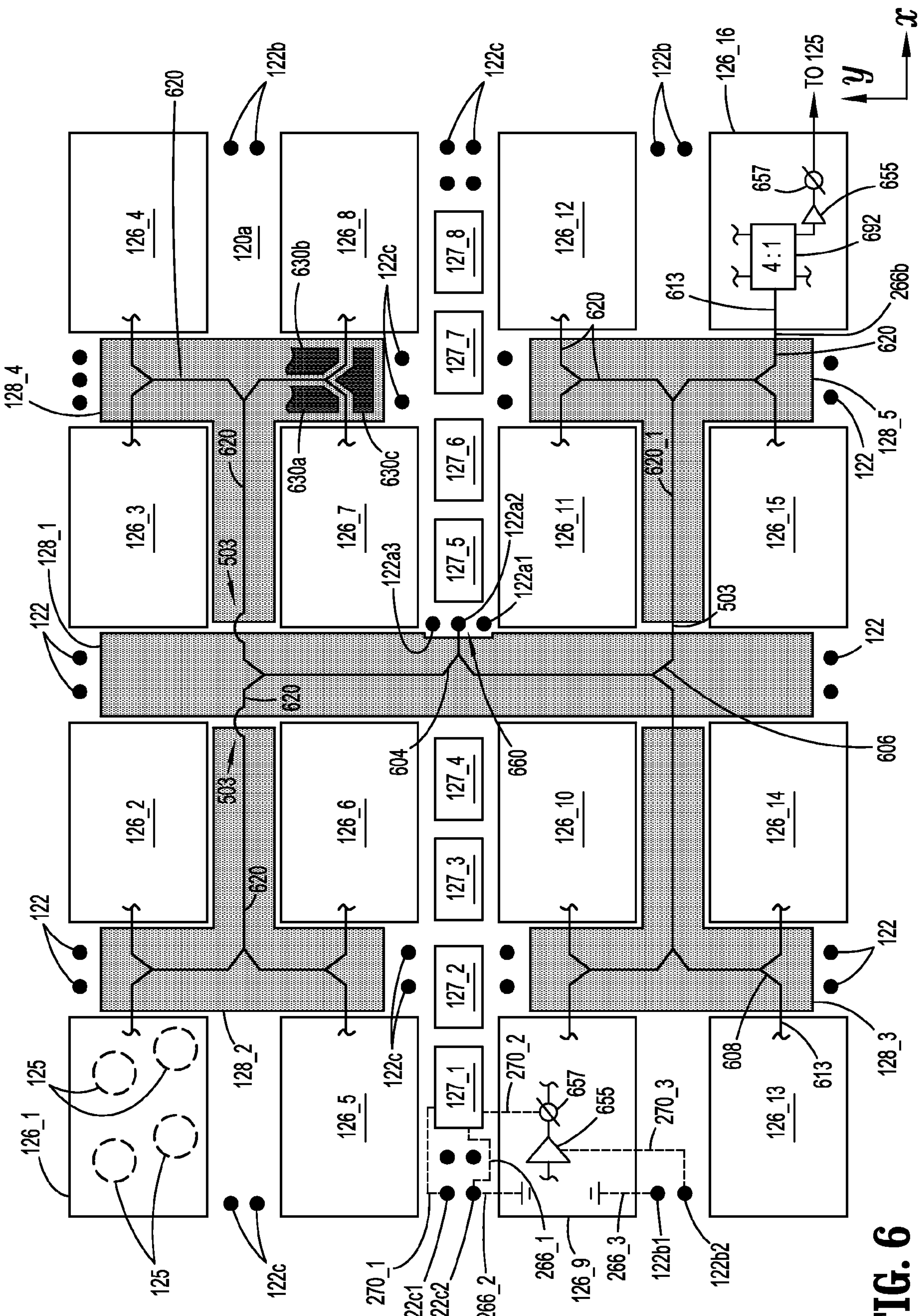
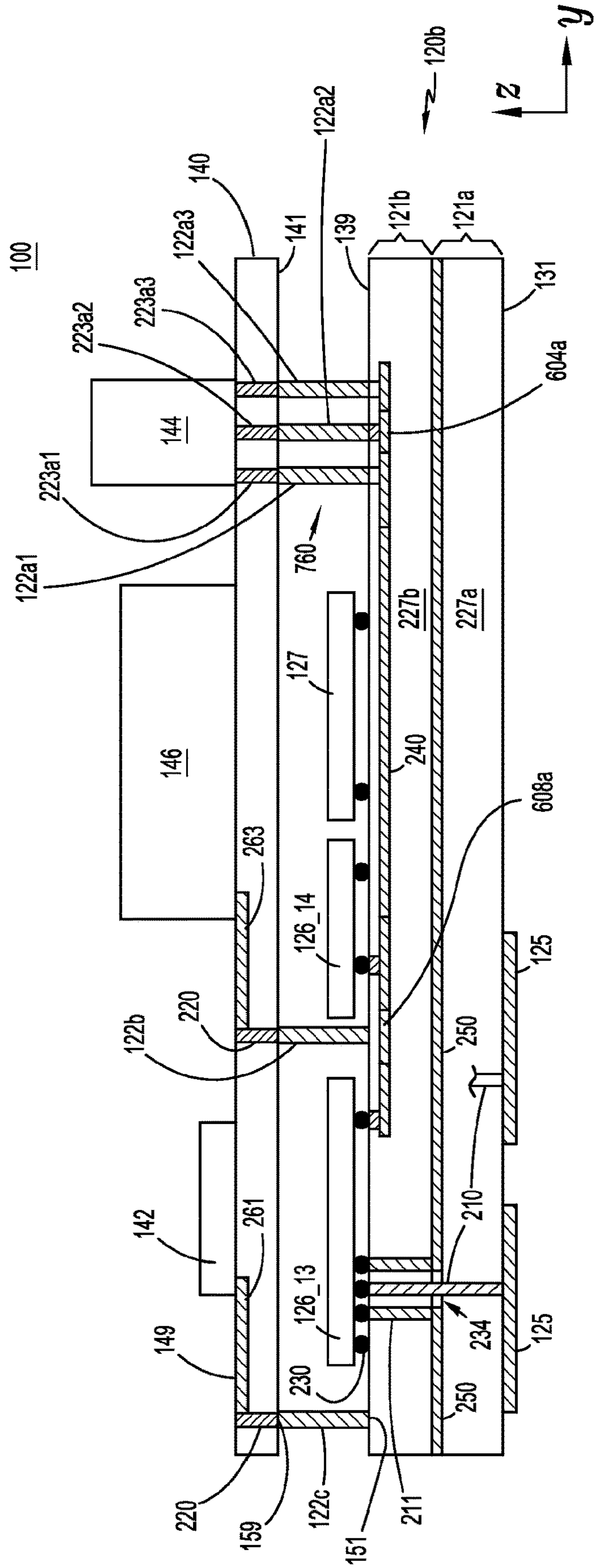

126_1
125
125
122c
128_2
126_5
122c
127_1
270_1
122c1
122c2
266_2
266_1
126_9
266_3
122b1
122b2
270_2
657
655
270_3
126_13
613
608
128_3
122
127_2
126_2
503
620
126_6
127_3
127_4
604
660
126_10
126_14
606
122
128_1
126_3
126_7
127_5
122a3
122a2
122a1
126_11
126_15
503
620_1
122
128_5
620
630a
630c
630b
128_4
126_4
120a
126_8
122c
127_6
127_7
127_8
126_12
126_16
613
4 : 1
692
657
655
266b
TO 125
122b
122c
y
x

FIG. 6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120b
y
z
140
141
139
121b
121a
131
122a2
122a3
223a3
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223a1
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144
604a
760
146
127
227b
227a
240
608a
263
126_14
125
220
122b
250
210
142
261
126_13
234
149
230
250
220
159
122c
151
211

FIG. 7

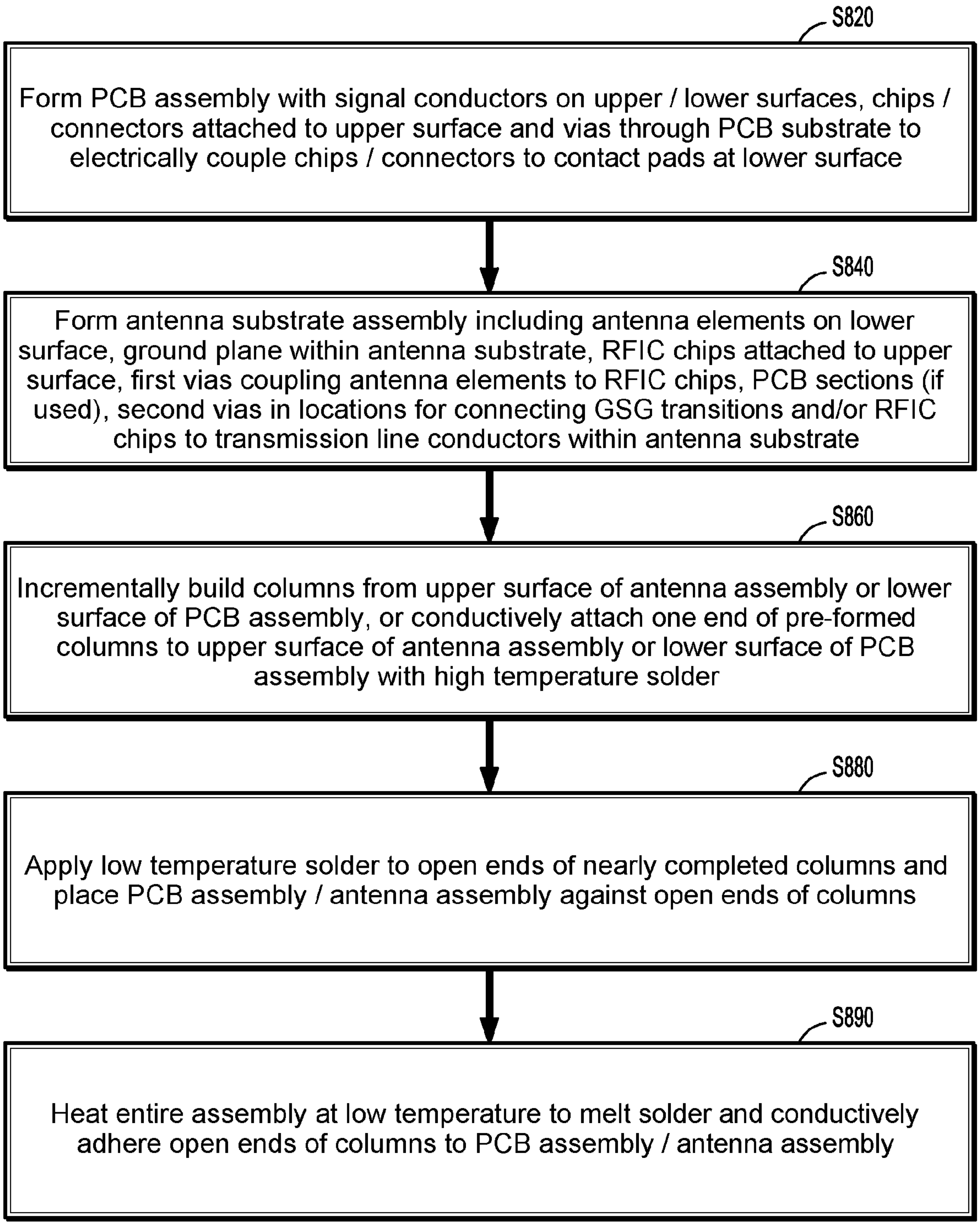
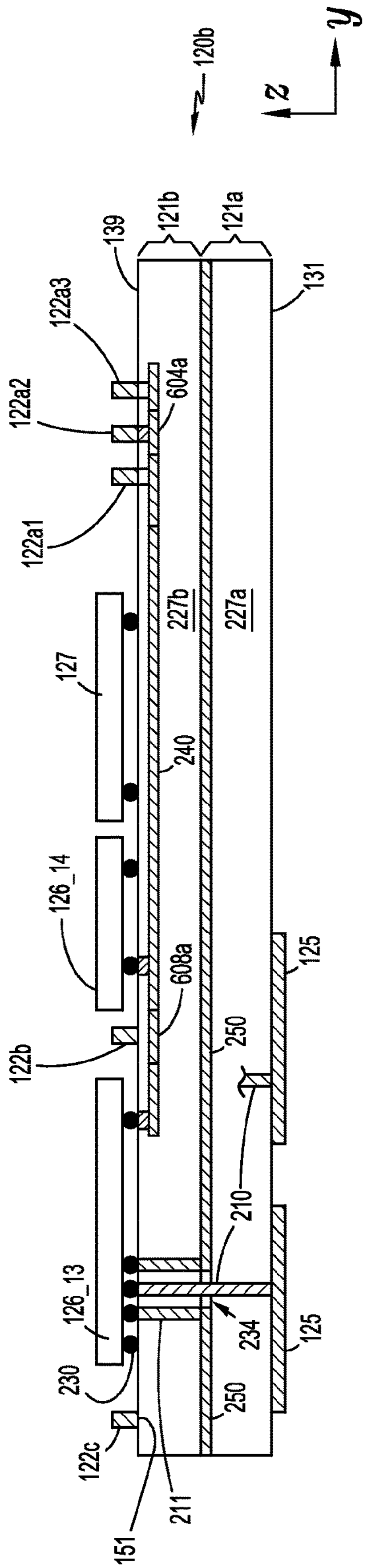

S820
Form PCB assembly with signal conductors on upper / lower surfaces, chips / connectors attached to upper surface and vias through PCB substrate to electrically couple chips / connectors to contact pads at lower surface
S840
Form antenna substrate assembly including antenna elements on lower surface, ground plane within antenna substrate, RFIC chips attached to upper surface, first vias coupling antenna elements to RFIC chips, PCB sections (if used), second vias in locations for connecting GSG transitions and/or RFIC chips to transmission line conductors within antenna substrate
S860
Incrementally build columns from upper surface of antenna assembly or lower surface of PCB assembly, or conductively attach one end of pre-formed columns to upper surface of antenna assembly or lower surface of PCB assembly with high temperature solder
S880
Apply low temperature solder to open ends of nearly completed columns and place PCB assembly / antenna assembly against open ends of columns
S890
Heat entire assembly at low temperature to melt solder and conductively adhere open ends of columns to PCB assembly / antenna assembly

FIG. 8



120b
y
z
139
121b
121a
131
122a3
122a2
604a
122a1
227b
227a
127
240
126_14
608a
125
122b
250
210
126_13
234
125
230
250
122c
211
151

FIG. 9A

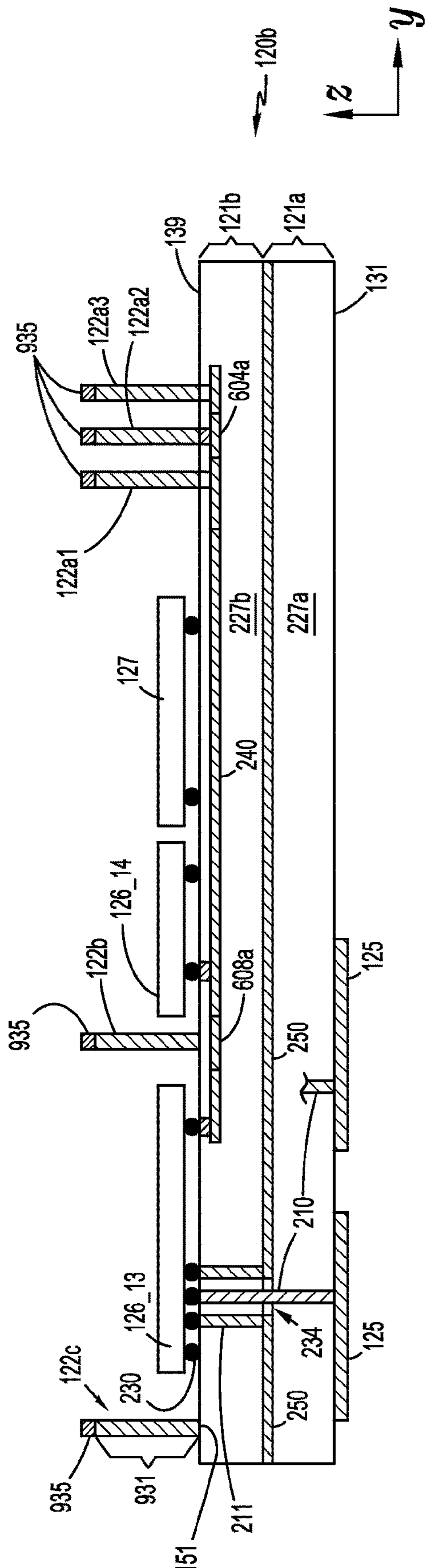
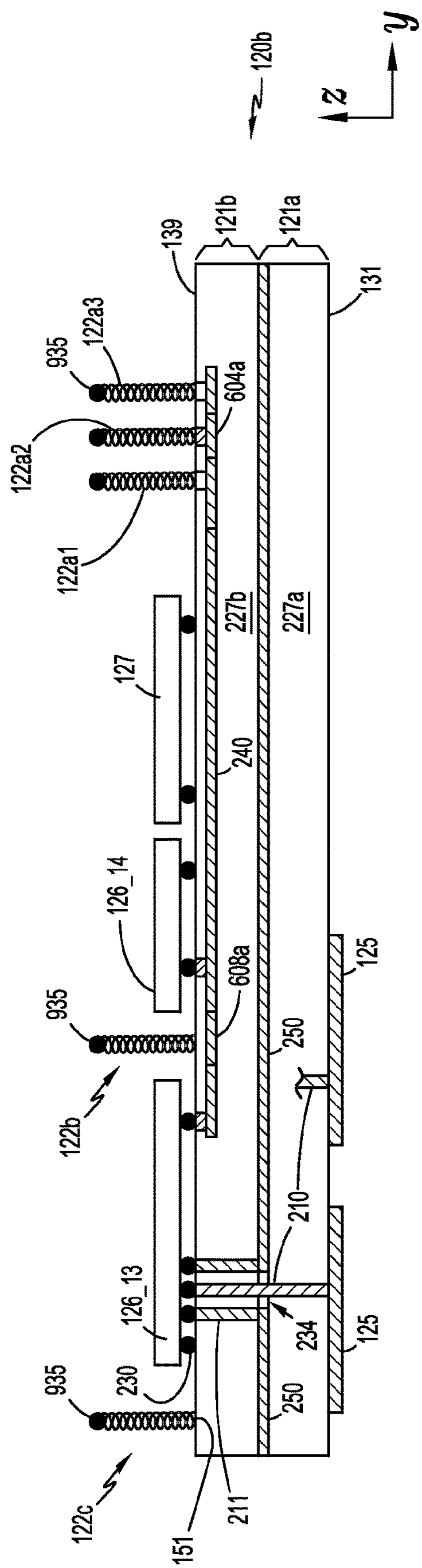

120b
z
y
139
121b
121a
131
935
122a3
122a2
604a
122a1
227b
227a
127
240
126_14
122b
935
608a
250
125
210
126_13
122c
230
234
125
250
935
931
151
211

FIG. 9B



120b
y
z
121b
121a
139
131
935
122a3
604a
122a2
122a1
227b
227a
127
240
126_14
608a
125
935
250
122b
210
126_13
234
125
230
250
935
122c
151
211

FIG. 9C

ANTENNA ARRAY ARCHITECTURE WITH ELECTRICALLY CONDUCTIVE COLUMNS BETWEEN SUBSTRATES

CROSS REFERENCE TO RELATED APPLICATION

The present Application is a 371 National Stage entry of PCT application no. PCT/US2020/050681, filed Sep. 14, 2020, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

This disclosure relates generally to compact architectures for antenna arrays integrated with radio frequency integrated circuit chips (RFICs).

DISCUSSION OF RELATED ART

Antenna arrays are deployed for a variety of applications at microwave and millimeter wave frequencies, e.g., in aircraft, satellites, vehicles, and base stations for general land-based communications. Such antenna arrays typically include microstrip radiating elements driven with phase shifting beamforming circuitry to generate a beam steerable phased array. It is typically desirable for an entire antenna system, including the antenna array and beamforming circuitry, to occupy minimal space with a low profile while meeting requisite performance metrics over a range of environmental conditions.

A low profile antenna array apparatus may be constructed with antenna elements integrated with RFICs (e.g. MMICs) in a compact structure. The antenna array apparatus may have a sandwich type configuration in which the antenna elements are disposed in an exterior facing component layer and the RFICs are distributed across the effective antenna aperture within a proximate, parallel component layer behind the antenna element layer. The RFICs may include RF power amplifiers (PAs) for transmit operations, low noise amplifiers (LNAs) for receive operations, and/or phase shifters/amplitude adjusters for beam steering. By distributing PAs/LNAs in this fashion, high efficiency on transmit and/or low noise performance on receive are realizable. Complex connection layouts may bring DC biasing voltages to the amplifiers and control signals for beam steering to the phase shifters. A typical antenna array apparatus for generating a narrow beamwidth of just a few degrees may include hundreds or thousands each of RFIC chips, antenna elements and control lines. Such a complex arrangement presents design challenges to produce a low profile design suitably operational in a wide range of environments.

SUMMARY

In an aspect of the present disclosure, an antenna apparatus includes an antenna substrate with opposite first and second surfaces; and a printed circuit board (PCB) having opposite first and second surfaces. A plurality of antenna elements are disposed at the first surface of the antenna substrate. A plurality of electrically conductive columns, each having a first end attached to the second surface of the PCB and a second end attached to the second surface of the antenna substrate, secure the PCB to the antenna substrate and provide an electrical interconnect between the PCB and the antenna substrate. A plurality of radio frequency integrated circuit (RFIC) chips are each attached to the second surface of the antenna substrate and are coupled to the plurality of antenna elements. At least one circuit element is attached to the first surface of the PCB and electrically coupled to at least one of the RFIC chips through at least one of the electrically conductive columns.

In various examples, the electrical interconnect can be an RF interconnect to communicate an RF signal; a DC interconnect to communicate a DC signal; or a data control signal interconnect to communicate a data control signal. The PCB and the antenna substrate can have different coefficients of thermal expansion (CTEs), and the columns deflect, flex or compress to provide relief of stress due to the different CTEs.

In another aspect, an interconnection structure for an electronic device includes a substrate having an upper surface; a PCB having an upper surface and a lower surface; and a plurality of electrically conductive columns attached between the lower surface of the PCB and the upper surface of the substrate. The columns secure the PCB to the substrate and provide electrical interconnects between the PCB and the substrate. A plurality of IC chips are attached to the upper surface of the substrate. At least one circuit element is attached to at least one of the upper surface and the lower surface of the PCB and is electrically coupled to at least one of the IC chips through at least one of the electrically conductive columns.

BRIEF DESCRIPTION OF THE DRAWINGS

The above and other aspects and features of the disclosed technology will become more apparent from the following detailed description, taken in conjunction with the accompanying drawings in which like reference characters indicate like elements or features. Various elements of the same or similar type may be distinguished by annexing the reference label with an underscore/dash and second label that distinguishes among the same/similar elements (e.g., _1, _2), or directly annexing the reference label with a second label. However, if a given description uses only the first reference label, it is applicable to any one of the same/similar elements having the same first reference label irrespective of the second label. Elements and features may not be drawn to scale in the drawings.

FIG. **1** is an exploded perspective view of an example antenna apparatus according to an embodiment.

FIG. **2** is an example cross-sectional view of a portion of the antenna apparatus of FIG. **1** according to an embodiment.

FIG. **3**A illustrates an example interconnect structure of the region “A” in FIG. **2**, in which an electrically conductive column is conductively adhered to a first metal layer of an antenna substrate.

FIG. **3**B illustrates another example interconnect structure of the region “A” in FIG. **2**A, in which an electrically conductive column is conductively adhered to a second metal layer of the antenna substrate.

FIG. **4** depicts an example interconnect structure of the region “B” in FIG. **2**, in which an electrically conductive column is conductively adhered to a transmission line layer of the antenna substrate.

FIG. **5**A is an example cross-sectional view of a portion of the antenna apparatus of FIG. **1** according to another embodiment.

FIG. **5**B is a cross-sectional view of an example structure of the region “C” in FIG. **5**A.

FIG. **6** is an example view taken along the line **6-6** of FIG. **5**A, and illustrates a layout of IC chips and PCB sections attached to a surface of the antenna substrate.

FIG. **7** is an example cross-sectional view of a portion of the antenna apparatus of FIG. **1** according to a further embodiment.

FIG. **8** is a flow chart illustrating an example method of fabricating an antenna apparatus according to an embodiment.

FIG. **9**A illustrates an antenna substrate assembly of the antenna apparatus with electrically conductive columns partially formed thereon.

FIG. **9**B illustrates the antenna substrate assembly with the columns fully formed thereon.

FIG. **9**C shows the antenna substrate assembly with pre-formed columns attached thereto.

DETAILED DESCRIPTION OF EMBODIMENTS

The following description, with reference to the accompanying drawings, is provided to assist in a comprehensive understanding of certain exemplary embodiments of the technology disclosed herein for illustrative purposes. The description includes various specific details to assist a person of ordinary skill the art with understanding the technology, but these details are to be regarded as merely illustrative. For the purposes of simplicity and clarity, descriptions of well-known functions and constructions may be omitted when their inclusion may obscure appreciation of the technology by a person of ordinary skill in the art.

FIG. **1** is an exploded perspective view of an example antenna apparatus, **100**, according to an embodiment. Antenna apparatus **100** has a compact arrangement that includes a plate-like antenna substrate **120** secured to a printed circuit board (PCB) **140** by a plurality of electrically conductive columns **122**. A plurality of antenna elements **125** are disposed at a first major surface **131** (in the x-y plane) of antenna substrate **120**. A plurality of radio frequency integrated circuit (RFIC) chips **126** are each attached to a second, opposite major surface **139** of antenna substrate **120** and are coupled to antenna elements **125**. At least one circuit element, such as a field programmable gate array (FPGA) **142**, a DC connector **146** and/or an RF connector **144**, are attached to a first major surface **149** of PCB **140**. Each of the columns **122** may provide an electrical interconnect between the circuit element(s) secured to PCB **140** and RFIC chips **126**, through conductive layers within antenna substrate **120**.

To provide the electrical interconnect, each of columns **122** may have a first end **151** attached to second surface **139** of antenna substrate **120** and a second end **159** attached to a second surface **141** of PCB **140**. Antenna substrate **120** and PCB **140** may have different coefficients of thermal expansion (CTEs). The structure and material of columns **122** may be designed to permit stress relief by allowing antenna apparatus **100** (hereafter, "antenna **100**" interchangeably) to accommodate CTE mismatch between antenna substrate **120** and PCB **140**. To this end, the column **122** configuration allows it to deform by flexing, compressing or stretching as the PCB and antenna substrate expand at different rates over temperature, while maintaining the electrical connection. This contrasts to a solder ball, which may break when subject to the same stress. For instance, columns **122** may each be configured with shapes such as a solid cylinder; a solid cylinder with a spiraling skin of a different material; a spring; a flexible solid structure; or a micro-coaxial cable section. Columns **122** may be composed of solder (e.g. a Pb/Sn alloy) or other conductive material. In one example, columns **122** are column grid array (CGA) type columns with a Pb/Sn alloy interior cylinder and a spiraling wrapped skin made of copper for better heat conduction and reliability. In FIG. **1**, only a few columns **122**, RFIC chips **126**, antenna elements **125**, etc. are shown for simplicity of illustration. In a typical embodiment of antenna **100**, antenna elements **125**, RFIC chips **126** and columns **122** may each number in the hundreds or thousands. Further, a plurality of FGPAs **142**, a plurality of DC connectors **146** and/or a plurality of RF connectors **144** may be included in antenna **100** and attached to PCB **140**. The entire mechanical support for the attachment of PCB **140** to antenna substrate **120** may be provided by columns **122**.

For instance, columns **122** may be formed as solid solder columns by a process that incrementally builds up the columns in layers of solder. An example process may first form a lowest layer of all columns **122** against a solder pad on the second surface **139** of antenna substrate or second surface **141** of PCB **140**. The lowest layer may be formed sequentially using a computer controlled solder tool that moves in a sequence from column to column. The process may be repeated layer by layer to incrementally build the height of the columns (in the z plane) until a desired height is reached on all columns **122**. The last solder layer may be composed of a lower temperature solder than the solder on the other layers of the columns **122**. For instance, if columns **122** are built up from the surface of antenna substrate **120**, a final adhering step may involve placing PCB **140** atop the columns **122** by aligning low temperature solder pads on second surface **141** of PCB **140** with the columns **122**. Then, the entire antenna assembly may be heated at a temperature sufficient to melt only the low temperature solder to complete the PCB **140** to antenna substrate **120** adhering process.

For example, PCB **140** may have one or more FPGAs **142**, DC connectors **146** and RF connectors **144** attached to its first surface **149**, and these circuit elements may electrically connect to signal and/or ground lines formed within antenna substrate **120** through columns such as **122***a*, **122***b* and **122***c*. Antenna substrate **120** may have multiple thin metal layers formed therein which may be patterned to form DC bias lines, control signal lines, ground lines, and RF transmission lines. For instance, column **122***b* may be a DC interconnect that couples DC connector **146** to a DC line within antenna substrate **120** to communicate a DC signal. The DC signal may bias amplifiers within RFIC chips **126**. Column **122***c* may be a data control signal interconnect that couples FPGA **142** to a control signal line within antenna substrate **120** to communicate a data control signal. The data control signal may be provided to RFIC chip **126** and/or other IC chips such as a serial peripheral interface (SPI) chip **127**. SPI chip **127** may generate phase shifter control signals based on the data control signal, which are provided to multiple RFIC chips **126** to adjust phase shifters therein and thereby operate antenna **100** as a phased array. Columns **122***a* may be RF interconnects that couple RF connector **144** to a transmission line to communicate an RF signal.

Antenna **100** may include a beamforming network (BFN) formed partially or entirely on at least one printed circuit board (PCB) section **128** (hereafter exemplified as a plurality of PCB sections **128**) coplanarly situated between RFIC chips **126** and attached to second surface **139** of antenna substrate **120**. Each PCB section **128** may be composed of a dielectric substrate such as alumina, a signal conductor, and at least one ground conductor, which collectively form a transmission line section, e.g., in coplanar waveguide

(CPW) or microstrip. The provision of multiple PCB sections **128** rather than a single PCB section **128** may facilitate manufacturing of antenna **100**. The BFN is a combiner/divider network, sometimes called a "corporate network" or a "distribution network", that divides an input RF transmit signal into N divided "element signals" for transmission by N respective antenna elements **125** forming the antenna array, and/or combines N receive path element signals provided by the N antenna elements **125** into a final composite receive signal. RF connector **144** may electrically connect to an input of the BFN (in the transmit case) and/or an output of the BFN (in the receive case). In another embodiment, the PCB sections **128** are omitted and the BFN is instead formed within a layer of antenna substrate **120**. Different ways of connecting the BFN to RF connector **144** will be discussed later.

Antenna elements **125** may each be a microstrip patch antenna element printed on antenna substrate **120** to form a planar array. Other types of antenna elements such as dipoles or monopoles may be substituted. Antenna elements **125** may be electrically or electromagnetically coupled to ("fed from") an RFIC chip **126** at a respective feed point. RFIC chips **126** may be mechanically connected to antenna substrate **25** by solder bump connections or the like to connection pads located on antenna substrate **25**. RFIC chips **126** may also be mechanically and electrically connected to an antenna ground plane proximately below surface **139** in a "single RF layer substrate" case discussed below. In a typical embodiment, each RFIC chip **126** is coupled to multiple (e.g. several) antenna elements **125**, and RFIC chips **126** are distributed across substantially the entire effective aperture of the planar array formed by antenna elements **125**.

Antenna elements **125**, when embodied as microstrip patches, may have any suitable shape such as circular, square, rectangular, elliptical or variations thereof, and may be fed and configured in a manner sufficient to achieve a desired polarization, e.g., circular, linear, or elliptical. The number of antenna elements **125**, their type, sizes, shapes, inter-element spacing, and the manner in which they are fed may be varied by design to achieve targeted performance metrics. In a typical embodiment antenna **100** may include hundreds or thousands of antenna elements **125**. In embodiments described below, each antenna element **125** is a microstrip patch fed with a probe feed. The probe feed may be implemented as a via that electrically connects to an input/output (I/O) pad of an RFIC chip **126**. An I/O pad is an interface that allows signals to come into or out of the RFIC chip **126**. In other examples, an electromagnetic feed mechanism is used instead of a via, where each antenna element **125** is excited from a respective feed point with near field energy.

Antenna **100** may be configured for operation over a millimeter (mm) wave frequency band, generally defined as a band within the 30 GHz to 300 GHz range. In other examples, antenna **100** operates in a microwave range from about 1 GHz to 30 GHz, or in a sub-microwave range below 1 GHz. Herein, a radio frequency (RF) signal denotes a signal with a frequency anywhere from below 1 GHz up to 300 GHz. It is noted that an RFIC chip configured to operate at microwave or millimeter wave frequencies is often referred to as a monolithic microwave integrated circuit (MMIC). A MMIC is typically composed of III-V semiconductor materials or other materials such as silicon-germanium (SiGe).

Each RFIC chip **126** may include active beamforming circuitry, e.g., an amplifier and/or a phase shifter used to adjust one or more signals communicated with a connected antenna element(s) **125**. In embodiments where RFIC chips **126** include dynamically controlled phase shifters, antenna **100** is operable as a phased array for transmit and/or receive operations. In a phased array embodiment, a beam formed by antenna **100** is steered to a desired beam pointing angle set primarily according to the phase shifts of the phase shifters. Additional amplitude adjustment within RFIC chips **126** may also be included to adjust the beam pattern. With RF front end amplifiers and/or phase shifters distributed across the effective aperture of the antenna array, antenna **100** may be referred to as an active antenna array. In some embodiments, antenna **100** operates as both a transmitting and receiving antenna system, and each RFIC chip **126** includes receive circuitry comprising at least one low noise amplifier (LNA) for amplifying a receive signal, and at least one power amplifier (PA) for amplifying a transmit signal. In this case, each RFIC chip **126** may include suitable transmit/receive (T/R) switching/filtering circuitry to enable bidirectional signal flow on shared resources. Antenna **100** is alternatively configured to operate only as a receive antenna system or only as a transmit antenna system, in which case each RFIC chip **126** may include an LNA but not a PA, or vice versa.

FIG. **2** is a cross-sectional view of a portion of antenna **100** of FIG. **1** in an assembled state, according to an embodiment. The cross-sectional view represents a non-linear section of antenna **100** shown in FIG. **1**, with some distal features of the view omitted for clarity. In this example, antenna substrate **120** is configured in a "dual RF layer" configuration comprised of a first layer **121***a* and a second layer **121***b*. Top layers of antenna substrate **120** at second surface **139** such as those in region "A" (described later but not shown in FIG. **2**) may include one or more thin metal layers for DC and/or control signal routing.

Layer **121***a* may include a first dielectric layer **227***a*, and a metal layer **250** serving as an antenna ground plane for reflecting signal energy transmitted/received by antenna elements **125**. Antenna elements **125** forming a planar array are disposed at first surface **131** of antenna substrate **120** (the lower surface of dielectric layer **227***a*). Each antenna element **125** may be coupled to a respective RFIC chip **126** through a first via **210** serving as a signal conductor of a probe feed. First via **210** traverses an opening **234** within ground plane **250**. Second vias **211** on opposite sides of first via **210** serve as ground conductors for the probe feed and make electrical connections between respective ground contacts of RFIC chip **126** and ground plane **250**. Second layer **121***b* includes a second dielectric layer **227***b* and another metal layer **240** which is patterned to form RF transmission line conductors between various connection points. Dielectric layers **227***a*, **227***b* may be composed of any suitable dielectric such as fused silica. Connections between metal layer **240** and connection points (e.g., I/O pads) on RFIC chips **126**, PCB sections **128**, IC chips **127** and any column **122** making an RF connection, may each be made through a respective short via **243** and an electrical connection joint **230** such as a solder ball or copper pillar. For example, a first portion **240***a* of second metal layer **240** is a transmission line conductor coupling an I/O pad **215** of RFIC chip **126** to a connection point of a PCB section **128**. A second portion **240***b* is a transmission line conductor coupling another connection point of PCB section **128** to a connection point of a "signal column" **122***a***2** discussed below.

Columns **122** may each have a height greater than the thickness of any one of the components **126**-**128** attached to antenna substrate **120**. Thus, an air gap may exist between the top surfaces of components **128** and PCB **140**. As discussed above, one way of forming each column **122** is by building up liquified metal such as solder one layer at a time, sequentially from column to column. Each column may be aligned with a respective connection pad on second surface **139** of antenna substrate **120**. An alternative process may use pre-formed conductive columns and conductively adhere the bottom and top surfaces of the pre-formed columns to connection pads at surfaces **139** and **141**, respectively, using a robotic tool or the like. Some examples of pre-formed columns **122** include solid cylinders, solid cylinders with a copper wrapped skin ("copper wrapped columns"), springs, and micro-coaxial cables. As mentioned, the structure and material of columns **122** may be sufficient to permit stress relief by accommodating CTE mismatch between antenna substrate **120** and PCB **140**. As PCB **140** and antenna substrate **120** expand at different rates over temperature, the column **122** configuration may prevent breakage by flexing, compressing or stretching, while maintaining the mechanical and electrical connections. Copper wrapped columns, for example, may still maintain an electrical connection even when cracked.

Vias **220** formed within PCB **140** may couple conductive lines at top surface **149** to top ends **159** of respective columns **122**. For instance, a via **220** electrically couples column **122*c*** to a connection point of FPGA **142** through a conductive line **261** at top surface **149**. In other examples, a via **220** may connect directly to the connection point of FPGA **142** if the layout permits. Bottom end **151** of column **122*c*** is coupled to an I/O pad of RFIC chip **126** through a conductive line within dielectric layer **120*b*** (e.g., a portion of layer **266** of FIG. **3**A discussed later) and an electrically conductive joint **230**. If column **122*c*** is designated for routing a control signal, a ground connection path for the control signal to a ground connection point of RFIC chip **126** at its lower surface may similarly be comprised of: another via **220** coupled to a ground connection point of FPGA **142** (directly below it or through another conductive line at surface **149**); another column **122**; another conductor within dielectric layer **120*b***; and another conductive joint **230**.

Similarly, DC connector **146** may provide a positive or negative voltage to RFIC chip **126** from an electrical contact at a lower surface of DC connector **146**, through a path including a signal conductor **263** at surface **149**, a via **220**, column **122*b***, a first metal layer within dielectric layer **227*b***, and a conductive joint **230** connected to an I/O pad of RFIC chip **126**. A similar path through another column **122** and a second metal layer within dielectric layer **227*b*** may provide a ground connection for the DC voltage between ground contacts of DC connector **146** and RFIC chip **126**.

RF connector **144** may be a coaxial or other type of connector electrically coupled to an RF input and/or output port of the BFN within PCB section **128**. For example, a ground-signal-ground (GSG) transition ("GSG interconnect") **260** may be used in the coupling path and may include signal column **122*a*2**, a first "ground column" **122*a*1** on one side of signal column **122*a*2**, and a second ground column **122*a*3** on the opposite side of signal column **122*a*2**. These columns may connect at their upper ends to a first ground via **223*a*1**, a signal via **223*a*2**, and a second ground via **223*a*3**, respectively, formed through PCB **140**. RF connector **144** may include an inner conductor which is connected to signal via **223*a*2**, and an outer conductor connected on one side to first ground via **223*a*1** and on an opposite side to signal via **223*a*3**. Note that other connection structures such as a "GS" scheme utilizing just one ground column, or a scheme with three or more ground columns, may be substituted in other embodiments.

Signal column **122*a*3** may be coupled to the RF input and/or output point of the BFN within PCB section **128** through a redistribution layer (RDL) interconnect within antenna substrate **120**. This interconnect may comprise a connection path including a via **243** (seen in FIG. **4**) connected to the lower end of signal column **122*a*3**, one end of transmission line conductor **240*b***, another via **243** on the opposite end of conductor **240*b***, and a connection joint **230** connecting the latter to PCB section **128**. Ground columns **122*a*1** and **122*a*2** may connect to a microstrip ground layer within dielectric layer **120*b*** or CPW ground conductors formed in metal layer **240**.

FIG. **3**A illustrates an example interconnect structure of the region "A" in FIG. **2**. In this example, an upper structure of antenna substrate **120** includes a first isolation layer **262** such as a polymer, e.g., Benzocyclobutene (BCB), the top surface of which forms the top surface **139** of antenna substrate **120**. A first metal layer **266** ("first conductive trace layer") is directly below first isolation layer **262** and may be designated for forming ground conductors for DC and/or control signals, or for forming signal conductors for the DC/control signals. A second metal layer **270** ("second conductive trace layer") is beneath first ground layer **266** and separated therefrom by a second isolation layer **268**. If first metal layer **266** is designated for ground conductors for the DC/control signals, second metal layer may be designated for forming the signal conductors for these signals, and vice versa. Metal layer **240*a*** used for transmission line conductors is situated between an upper portion of dielectric layer **227*b*** (directly beneath second metal layer **270**) and a lower portion of dielectric layer **227*b***.

In the example of FIG. **3**A, column **122*b*** is electrically connected to first layer **266** through an opening in isolation layer **262** slightly larger than the diameter of column **122*b***. To facilitate formation of an electrical connection joint, a surface finish metal layer **265** such as Electroless Palladium Immersion Gold (ENEPIG) or a nickel/gold alloy may have been formed within the opening in isolation layer **262**. Layer **265** may have been deposited to have a base portion atop metal layer **266**, a peripheral wall portion around the periphery of the opening, and an annular ring region atop surface **139**, to form a cavity. A well of solder or other liquefiable metal **264** may fill the cavity, adhering to both the surface finish layer **265** and the lower end of column **122*b*** to form the mechanical connection between column **122*b*** and antenna substrate **120** and the electrical connection to metal layer **266** therein. In other embodiments, surface finish metal layer **265** is omitted.

Each of metal layers **266** and **270** and isolation layers **262** and **268** may be at least one order of magnitude thinner than the thickness of substrate **120**. For instance, each of these layers may have a thickness on the order of 2-10 μm whereas substrate **120** may be on the order of 250 μm thick. Metal layers **266** and **268** may each form signal/ground lines in the x-y plane having a width on the order of 12 μm and spaced from one another by a spacing on the order of 12 μm. Each of layers **266** and **268** may have been etched or otherwise patterned to form hundreds or thousands of signal lines and ground lines in a typical embodiment of antenna **100**.

FIG. **3**B illustrates another example interconnect structure of the region "A" in FIG. **2**. In this example, column **122*b*** is electrically connected to second metal layer **270** through openings in first isolation layer **262**, first metal layer **266** and second isolation layer **268**. The openings in these layers may have been formed by aligning resist material with different geometries layer by layer during deposition of the respective layers. To prevent column **122***b* from shorting to first metal layer **266**, first metal layer **266** may been formed by deposition patterning with a larger opening than those of first and second isolation layers **268** and **262**. An annular isolation region **287** may have been formed at the depth of metal layer **266** to isolate first metal layer **266** from a subsequent electrical connection between column **122***b* and second metal layer **270**. A surface finish layer **285** akin to surface finish layer **265** may have been formed using electroplating or the like. Surface finish layer **285** may have a base portion on second metal layer **270**, annular wall portions against the edges of isolation layers **262**, **266** and **268** in the respective openings, and a rim portion on upper surface **139**. This results in a metal-lined cavity which can be filled with solder or other liquefiable metal **284**. When the solder **284** cools while the lower end of column **122***b* is placed in the cavity, the solder **284** electrically connects column **122***b* to second layer **270** through surface finish layer **285**. In other embodiments, surface finish layer **285** is omitted.

FIG. **4** illustrates an example interconnect structure of the region "B" in FIG. **2**, in which signal column **122***a***2** is conductively adhered to transmission line conductor **240***b* through via **243** within antenna substrate **120**. Via **243** may connect on its upper end to a disc-shaped catch pad **270***a* formed within metal layer **270**. For instance, metal layer **270** may have been formed atop dielectric layer **227***b* prior to forming via **243**. When forming metal layer **270**, catch pad **270***a* may have been formed by concentrically aligning a ring-shaped resist material with a circular region of via **243** (to be formed subsequently). Metal layer **270** may have then been deposited, resulting in a ring-shaped opening around catch pad **270***a*. Via **243** may have next been formed through catch pad **270***a*. Isolation material may have been deposited in a subsequent step to form an annular isolation region **289** within the openings around catch pad **270***a*, thereby isolating the remaining material of metal layer **270** from via **243**. An interconnect structure between catch pad **270***a* and the lower end of column **122***a***2** may be the same as that described in connection with FIG. **3**B for the connection to column **122***b*. That is, the interconnect structure may comprise surface metal layer **285** with a base portion atop catch pad **270***a* and annular wall portions and a rim portion, forming a cavity which is filled with liquifiable metal **284** as described above.

Transmission line conductor **240***b* may be e.g. a microstrip conductor, a coplanar waveguide (CPW) conductor, or a stripline conductor. If transmission line conductor **240***b* is configured as a CPW conductor, ground columns **122***a***1** and **122***a***3** may be conductively adhered to other respective portions of transmission line layer **240** on opposite sides of layer portion **240***b* in the same manner as described for FIG. **4**. In the case of microstrip, the ground plane for the microstrip can be a region of the second metal layer **268** overlaying conductor **240***b*. In this case, ground columns **122***a***1** and **122***a***3** may each be conductively adhered to second metal layer **270** in the same fashion as described above for FIG. **3**B. Alternatively, a region of first metal layer **266** may be used as the ground plane (by removing the corresponding region of second metal layer **270**), in which case ground columns **122***a***1** and **122***a***3** may be conductively adhered to first metal layer **266** in the same fashion as was described for FIG. **3**A. In the case of stripline, an additional metal layer would be provided beneath layer **240***b* and electrically connected to the ground plane provided by first or second metal layers **266**, **270**.

FIG. **5**A is an example cross-sectional view of a portion of the antenna apparatus of FIG. **1** according to another embodiment. (The view of FIG. **5**A represents a different non-linear slice of antenna **100** than that of FIG. **2**.) This embodiment employs a "single RF layer" antenna substrate **120***a* (another embodiment of antenna substrate **120** of FIG. **1**) in which an antenna ground plane **550** is located within antenna substrate **120***a* proximate its top surface **139**. Note that the cross-section of FIG. **5**A shows signal column **122***a***2** electrically coupled between RF connector **144** and antenna substrate **120***a* through via **223***a***2**. In this view, it may be assumed that first and second ground columns **122***a***1** and **122***a***3** are not visible since they are respectively behind, and in front of, signal column **122***a***2** (or vice versa).

FIG. **5**B is a cross-sectional view of an example structure of the region C in FIG. **5**A. Antenna substrate **120***a* may be composed of dielectric layer **527** and alternating metal/isolation layers at the upper portion of substrate **120***a*. These may include antenna ground plane **550** directly atop dielectric layer **527**, which is sequentially followed towards top surface **139** by an isolation layer **540**, second metal layer **270**, isolation layer **268**, first metal layer **266** and uppermost isolation layer **262**.

Signal column **122***a***2** may conductively adhere to a first end of a conductive trace **266***a* formed in metal layer **266** through a solder well **264** and a surface finish metal layer **265**, akin to those described above in connection with FIG. **3**A. Conductive trace **266***a* may be isolated from adjacent portions of first metal layer **266** by a ring-shaped isolation region **595***a* surrounding conductive trace **266***a*. An electrical interconnect may be formed between a second, opposite end of conductive trace **266***a* and a CPW or microstrip signal conductor **604** at the top surface of PCB section **128**. This interconnect may include a via **593** formed within PCB section **128**, an electrical connection joint **230**, and a surface finish metal layer **585** formed on the second end of conductive trace **266***a*. An isolation section **262***a* of isolation layer **262** may have been formed to support walls of surface finish metal layer **265**. Alternatively, the section **265***a* is substituted with a conductive layer section, e.g., additional surface finish metal layer material.

If PCB section **128** is configured as a CPW transmission line, first and second ground conductors on opposite sides of signal conductor **604** may connect to lower ends of ground columns **122***a***1** and **122***a***3**, respectively, using a similar configuration as that of FIG. **5**B. In this case, the respective connections may be made through first and second additional conductive traces ("ground traces") formed in metal layer **266** on opposite sides of, and isolated from, conductive trace **266***a*. In other words, the first and second ground traces, in conjunction with conductive trace **266***a*, form an interconnecting section of CPW transmission line. Alternatively, the ground connections are made in a different metal layer within substrate **120***a* than the metal layer (e.g. **266**) used for the signal conductor connection.

If PCB section **128** is configured as a microstrip transmission line, the signal line **604** to signal column **122***b* connection may be made the same way as in FIG. **5**B, through a similar conductive trace **266***a*. In this case, a microstrip ground plane may be present at the lower surface of PCB section **128** and the lower end of via **593** may pass through an opening in the microstrip ground plane. Further, first and second ground traces may extend on opposite sides of conductive trace **266***a* and may each be connected on one end to a respective connection point of the microstrip ground plane, and on the opposite end to ground column **122***a***1** or **122***a***3**, respectively.

It is noted here that a similar interconnect as that illustrated in FIG. **5**B may be made between an I/O pad at a lower surface of an RFIC chip **126** and first metal layer **266**. (The interconnect may comprise a conductive joint **230** within a cavity lined with a surface finish metal layer **585**.) To connect any I/O pad of an RFIC chip **126** to a signal line of second metal layer **270**, openings are formed within each of first metal layer **266** and second isolation layer **268**. Peripheral surfaces of these openings may be lined with surface finish metal layer **585** extended to second metal layer **270** to form a cavity. The cavity may be filled with liquefiable metal to form a conductive well, akin to the conductive well **284** of FIG. **3**B. The conductive well may connect to the lower end of the I/O pad of RFIC chip **126**.

Antenna ground plane **550** may be electrically connected to a microstrip ground plane at the lower surface of RFIC chip **126** and/or PCB section **128**. The interconnect for this connection may comprise at least one electrical connection joint **230** and a solder well or the like within antenna substrate **120***a* extending to antenna ground plane **550**. Such a solder well may have an upper portion similar to solder well **284** of FIG. **3**B described above and a lower portion extending through an isolated opening in second metal layer **270** and connecting to ground plane **550**.

FIG. **6** is an example view taken along the line **6-6** of FIG. **5**A. This view illustrates an exemplary layout of IC chips and PCB sections looking down towards top surface **139** of antenna substrate **120***a*. Electrically conductive columns **122** (including columns **122***a*, **122***b*, **122***c*, etc.) may be distributed in both peripheral and inner regions of the layout. An example is presented illustrating **16** RFIC chips **126_1** to **126_16** arranged uniformly in rows and columns; eight serial peripheral interface (SPI) chips **127_***a* to **127_8** linearly arranged between a pair of adjacent rows of the RFIC chips **126**; and five PCB sections **128_***a* to **128_5** within which at least a portion of the BFN is formed. PCB sections **128** include signal conductors **620** of the BFN at its top surface, which may be electrically connected to respective signal conductors **613** within RFIC chips **126** through RDL interconnects **266***b* formed in first metal layer **266**, or interconnects formed in second metal layer **270**. (RDL interconnects **266***b* may be assumed visible in the view of FIG. **6** through first isolation layer **262**.) If PCB sections **128** are implemented as CPW, CPW ground conductors such as **630***a*, **630***b*, **630***c* are present on opposite sides of signal conductors **620**. These ground conductors **630***a*-**630***c* may each be electrically connected through RDL interconnects to CPW ground conductors (not shown) of RFIC chips **126** on opposite sides of signal conductors **613**. In the case of microstrip, a microstrip ground plane may be present at the lower surfaces of PCB sections **128** and RFIC chips **126**, which are suitably connected together. It is noted here that a similar layout as shown in FIG. **6** may be implemented for the embodiment of FIG. **2**; however, in this case the electrical connections between signal conductors **613** and **620** may be made through a transmission line layer (metal layer **240**) within antenna substrate **120**.

The BFN within PCB sections **128** may, in the transmit direction of antenna **100**, receive an input RF signal from RF connector **144** through a GSG transition **660** formed by signal column **122***a***2**, ground column **122***a***1** and ground column **122***a***3**. The BFN may include a 2:1 I/O coupler **604** having an I/O port electrically connected to GSG transition **660** through an RDL connection within antenna substrate **120***a* or **120** (e.g. through conductive traces within first and/or second metal layers **266**, **270** as described earlier). I/O coupler **604**, in conjunction with other couplers of the BFN such as **606** and **608** may divide the input RF signal into sixteen divided transmit signals which are provided to RFIC chips **126_1** to **126_16**, respectively. In the receive direction, RF receive signals output from RFIC chips **126_1** to **126_16** may be combined by the BFN and the combined receive signal is output to RF connector **144** through GSG transition **660**. Some examples of the BFN couplers **604**, **606**, etc. include Wilkinson dividers (e.g., with printed resistors between divided output lines); hybrid ring ("rat race") couplers; and 90° branch line couplers. In other layout examples, to form a 1:K divider/combiner within transmission line sections **128**, where K is other than 16, a larger or smaller number of 2:1 couplers and/or M:1 couplers (where M>2) are provided.

Each RFIC chip **126** may include active RF front end components such as at least one amplifier **655** and at least one phase shifter **657**. Antenna elements **125** coupled to an RFIC chip **126** may approximately overlay the RFIC chip **126**. For example, RFIC chip **126_1** is shown overlaying four antenna elements **125**; each of these antenna elements may be connected to a respective signal path including one amplifier **655** and one phase shifter **657**, which individually control amplitude and phase of signals traversing that antenna element **125**. (For clarity of illustration, one amplifier **655** and one phase shifter **657** are shown just in RFIC chips **126_9** and **126_16**. If, for example, four antenna elements **125** are coupled to each RFIC chip **126**, each RFIC chip **126** may include four amplifiers **655** and four phase shifters **657**.)

DC voltages and control voltages originating from DC connector **146** and FGPA **142** may be routed through columns **122** to RFIC chips **126** and SPI chips **127** via conductors formed in metal layers **266** and **270**. DC voltages may bias amplifiers **655** and/or be used by other circuitry within RFIC chips **126** and SPI chips **127**. For instance, a DC biasing voltage from DC connector **146** may be carried on column **122***b***2** and routed to amplifier **655** within RFIC chip **126_9** through conductive line **270_3** and an I/O contact pad at the lower surface of RFIC chip **126_9**. A ground return for the DC biasing voltage back to DC connector **146** may include a conductive line **266_3** and column **122***b***1**.

In a similar fashion, control signals may be routed from FGPA **142** to SPI chips **127** and/or RFIC chips **126** through columns **122***c*. For instance, phase/amplitude control signals for beam steering may be generated by FGPA **142** and routed to SPI chips **127**. As an example, a control signal generated by FGPA **142** may be applied across columns **122***c***1** and **122***c***2**. The control signal may be routed to SPI chip **127_1** across conductive lines **270_1** and **266_1**, connected to columns **122***c***1** and **122***c***2**, respectively. Based on the control signal, SPI chip **127_1** may generate a phase shifter control voltage and apply the same to phase shifter **657** through conductive line **270_2** to dynamically set a phase shift for signals traversing an antenna element **125** coupled to that phase shifter **657**. In an example, column **122***c***2** may provide a ground return path for various control signals via electrical connection to each of: a ground contact of FGPA **142**; a ground contact of SPI chip **127_1** (through conductive line **266_1**); and a ground contact of RFIC chip **126_9** (through conductive line **266_2**).

It is noted here that the BFN may be formed partially on PCB sections **128** and partially on RFIC chips **126**. The BFN may be considered the signal paths and the divider/combiner circuitry between GSG transition **660** and the feed points of the amplifier elements **125**. Thus, for example, if each RFIC chip **126** feeds N>1 antenna elements **125** as illustrated, there may be a N:1 combiner/divider **692** (N=4 in the example of FIG. **6**) within each RFIC chip **126**. The N:1 combiner/divider **692** may be situated between an input point **613** of the RFIC **126** and N inputs of N respective amplifiers **655** and/or phase shifters **657** each coupled to a respective antenna element **125**. The N:1 combiner/divider **692** may be considered an "early stage(s)" of the BFN, whereas the BFN portion composed of couplers **608**, **606**, etc. may be considered "later stages" of the BFN.

In the example layout of FIG. **6**, signal conductors **620** of adjacent PCB sections **128** may be connected to one another by wirebonds **503**. For CPW transmission lines, a pair of ground conductors akin to **630***a* and **630***c* on opposite sides of signal conductors **620** may likewise be connected to corresponding ground conductors in adjacent PCB sections **128** via respective wirebonds **503**. In an alternative embodiment, a common dielectric substrate is used for all of PCB sections **128_1** to **128_5** and wirebonds **503** are omitted.

In another embodiment, direct connections between adjacent PCB sections **128** are avoided by connecting BFN signal paths through RFIC chips **126** between different PCB sections **128**.

FIG. **7** is an example cross-sectional view of a portion of antenna **100** of FIG. **1** according to another embodiment. In this example, the beamforming network (BFN) is formed within an antenna substrate **120***b* rather than within PCB sections **128**, and PCB sections **128** are omitted. This embodiment employs a dual RF layer structure for antenna substrate **120***b*, similar to that described above for the embodiment of FIG. **2**. For example, RF connector **144** may be coupled to a CPW transmission line formed by metal layer **240** through a GSG transition **760** comprised of columns **122***a***1**, **122***a***2** and **122***a***3**. Alternatively, if metal layer **240** is used to form a microstrip signal conductor, ground columns **122***a***1** and **122***a***3** may be conductively adhered to another metal layer serving as a microstrip ground within antenna substrate **120***b*.

GSG transition **760** may be arranged at a location proximate to an I/O RF coupler **604***a* akin to I/O coupler **604** of FIG. **6**. Metal layer **240** may be patterned to form a BFN with a layout similar to that of FIG. **6** in one example. In this case, one signal path of the BFN may lead to a 2:1 directional coupler **608***a* (akin to coupler **608**), which splits an RF transmit signal into two divided signals that are applied to RFIC chips **126_13** and **126_14**. Reciprocal signal flow from RFIC chips **226** through the directional couplers to GSG transition **760** may occur for a receiving antenna system. Other aspects of this configuration may be the same as those described in connection with FIG. **2**.

FIG. **8** is a flow chart describing an example method of fabricating antenna apparatus **100**. A process step **820** may involve forming a PCB assembly **140** with signal conductors **261** on upper surface **149** and/or lower surface **141** (see e.g., FIG. **2**); with chips/connectors (e.g., **142**, **144**, **146**) attached to upper surface **149** and vias **220** through the PCB **140**'s substrate. The vias are formed to electrically couple the chips/connectors to contact pads at lower surface **141**.

A separate process step S**840** may involve forming an antenna substrate assembly including: antenna elements **125** on a lower surface **131** of an antenna substrate **120**; a ground plane (e.g. **250**, **550**) within antenna substrate **120**; first vias **210** for coupling antenna elements **125** to RFIC chips **126**; second vias **243** in locations for connecting GSG transitions (formed by adjacent columns **122***a***1**, **122***a***2**, **122***a***3**) and/or second vias **243** for connecting RFIC chips **126** to transmission line conductors within the antenna substrate; RFIC chips **126** and other chips **127** attached to the upper surface **139** of antenna substrate **120**; and PCB sections **128** (if used) attached to upper surface **139**.

With the PCB assembly and antenna substrate assembly separately formed, columns **122** may be formed and attached at one end to either the upper surface of antenna assembly or the lower surface of the PCB assembly (S**860**). One example technique for this process involves incrementally building the columns from upper surface **139** of antenna substrate **120** or from the lower surface of the PCB assembly. One layer of each column **122** may be formed at a time using a computer controlled solder tool that moves in a sequence from column to column and deposits a small amount of solder to incrementally build up each column. FIG. **9**A, for example, illustrates an interim assembly structure in which lower portions of columns **122***b*, **122***c*, etc. have been formed on upper surface **139** of antenna substrate **120**. The process may be repeated layer by layer to incrementally build the height of the columns until a desired height is reached on all columns **122**. A low temperature solder may be applied (S**880**) to the open ends of the nearly completed columns **122**. For example, as shown in FIG. **9**B, completed columns **122** may each have a majority portion **931** composed of high temperature solder and an end portion **935** composed of low temperature solder. Alternatively or additionally, a low temperature solder pad is formed at the lower surface of PCB **140** at locations aligned with columns **122**.

An alternative implementation for process S**860** involves conductively adhering pre-formed columns **122** to antenna substrate **120** or PCB **140**. For example, FIG. **9**C shows the antenna substrate assembly with pre-formed columns **122** in the shape of springs attached thereto. Lower ends of the springs may be attached to surface **139** of antenna substrate **120** with high temperature solder. Low temperature solder **935** may be applied to the upper ends of the springs after the lower ends are attached. As mentioned earlier, pre-formed columns **122** may have other configurations such as a solid Pb/Sn alloy interior cylinder and a spiraling wrapped skin made of copper for better heat conduction and reliability. High and low temperature solder may be applied to opposite ends of these other column configurations in the same or similar way.

Note that process step S**860** of forming/attaching the columns on the antenna substrate or PCB may be performed prior to the attachment of the IC chips/connectors on the antenna substrate or PCB of which the columns are initially attached.

With columns **122** thus conductively adhered at one end to the antenna assembly or the PCB assembly, and the PCB assembly or antenna assembly placed against the open ends of the columns **122**, the entire assembly may then be heated (S**890**) at a low temperature sufficient to melt just the low temperature solder. When the low temperature solder cools, the previously open ends of columns **122** are conductively adhered to conductive contacts of the PCB or antenna assembly.

The above-described embodiments have been described in the context of antenna apparatus **100**. Other implementations of the technology herein may be applied to non-antenna applications. For instance, in other electronic devices, antenna elements **125** are substituted with at least one other type of first circuit elements, such as IC chips. The RFIC chips **126** may be substituted with other types of second IC chips which are electrically coupled to the first IC chips through vias **210** extending through substrate **120**. Electrically conductive columns **122** of the electronic device may connect in the same way to at least one upper metal layer within substrate **120** to provide DC interconnects, control signal interconnects and/or RF signal interconnects. The column interconnects may connect third IC chips/connectors/components attached to the upper surface of PCB **140** to the second IC chips attached to upper surface **139** of substrate **120**. The resulting electronic device is formed in a compact three dimensional stacked structure. Further, if the substrate **120** and the PCB **140** of the electronic device have different coefficients of thermal expansion (CTEs), the same advantages of permitting stress relief between substrate **120** and PCB **140** as described above are applicable to the electronic device. That is, the structure and material of columns **122** may be sufficient to permit stress relief by accommodating CTE mismatch between substrate **120** and PCB **140**. As PCB **140** and substrate **120** expand at different rates over temperature, the column **122** configuration may prevent breakage by flexing, compressing or stretching, while maintaining the mechanical and electrical connections.

Moreover, in the above-described embodiments, chip(s) **142** DC connector(s) **146** and/or RF connector(s) **144** are shown and described attached to an upper surface **149** of PCB substrate **140**. In other embodiments, if space is available between antenna substrate **120** and PCB **140** and a flat, component-free surface is desirable at the upper surface **149**, chips/connectors may be alternatively adhered to the lower surface of PCB **140**. In this case, connectors such as **144**, **146** may have side-facing connection ports. In these or other embodiments, no circuit elements or antenna elements may be provided at the lower surface **131** of substrate **120**.

While the technology described herein has been particularly shown and described with reference to example embodiments thereof, it will be understood by those of ordinary skill in the art that various changes in form and details may be made therein without departing from the spirit and scope of the claimed subject matter as defined by the following claims and their equivalents.

What is claimed is:

1. An antenna apparatus comprising:
- an antenna substrate having opposite first and second surfaces;
- a plurality of antenna elements disposed at the first surface of the antenna substrate;
- a printed circuit board (PCB) having opposite first and second surfaces;
- a plurality of deformable electrically conductive columns, each comprising at least one of springs, micro-coaxial cables, and cylinders with copper spiral wrappings, and each having a first end attached to the second surface of the PCB and a second end attached to the second surface of the antenna substrate, to secure the PCB to the antenna substrate and to provide an electrical interconnects between the PCB and the antenna substrate;
- a plurality of radio frequency integrated circuit (RFIC) chips each attached to the second surface of the antenna substrate and coupled to the plurality of antenna elements; and
- at least one circuit element attached to the first surface of the PCB and electrically coupled to at least one of the RFIC chips through at least one of the electrically conductive columns.

2. The antenna apparatus of claim **1**, wherein at least one of the electrical interconnects is a radio frequency (RF) interconnect to communicate an RF signal.

3. The antenna apparatus of claim **1**, wherein at least one of the electrical interconnects is a direct current (DC) interconnect to communicate a DC signal.

4. The antenna apparatus of claim **1**, wherein at least one of the electrical interconnects is a data control signal interconnect to communicate a data control signal.

5. The antenna apparatus of claim **4**, wherein the data control signal is provided to multiple ones of the RFIC chips to adjust phase shifters therein to create a phased array.

6. The antenna apparatus of claim **1**, wherein the PCB and the antenna substrate have different coefficients of thermal expansion (CTEs), and the columns are configured to deform by flexing, compressing or stretching to provide relief of stress due to the different CTEs.

7. The antenna apparatus of claim **1**, wherein the at least one circuit element comprises an IC chip that provides control signals and/or bias voltages to the RFIC chips through the at least one of the columns and a conductive trace layer at the second surface of the antenna substrate.

8. The antenna apparatus of claim **7** wherein the IC chip is a field-programmable gate array (FPGA).

9. The antenna apparatus of claim **1**, wherein the at least one circuit element comprises a DC connector that provides a DC voltage to the RFIC chips through the at least one of the columns and a conductive trace layer at the second surface of the antenna substrate.

10. The antenna apparatus of claim **1**, wherein the at least one circuit element comprises an RF connector that routes an RF signal to or from the RFIC chips through the at least one of the columns.

11. The antenna apparatus of claim **10**, wherein:
- the plurality of columns comprise a first column and second and third columns on opposite sides of the first column; and
- the RF connector is coupled to the RFIC chips through a ground-signal-ground interconnect comprising the first column serving as a signal interconnect and the second and third columns serving as respective ground interconnects.

12. The antenna apparatus of claim **1**, wherein:
- the second surface of the antenna substrate comprises an antenna ground plane having openings for interconnects to electrically couple the RFICs and the antenna elements; and
- the RFIC chips have respective ground connection points electrically coupled to the antenna ground plane.

13. The antenna apparatus of claim **1**, wherein each of the RFIC chips has a lower surface attached to the second surface of the antenna substrate through a plurality of electrical connection joints, and an upper surface separated from the second surface of the PCB by an air gap.

14. The antenna apparatus of claim **13**, wherein the plurality of electrical connection joints are solder bumps or copper pillars.

15. The antenna apparatus of claim **1**, wherein the plurality of electrically conductive columns comprise the springs.

16. The antenna apparatus of claim **1**, wherein;
- the plurality of electrically conductive columns comprise the cylinders with copper spiral wrappings;
- the cylinders are solder columns; and
- the electrically conductive columns form a column grid array (CGA).

17. The antenna apparatus of claim **1**, wherein the RFIC chips each comprise at least one of a receive amplifier, a transmit amplifier and a phase shifter.

18. The antenna apparatus of claim **1**, wherein the RFIC chips are each coupled to one or more of the antenna elements through a respective via formed within the antenna substrate.

19. The antenna apparatus of claim **1**, further comprising a beamforming network (BFN) formed on at least one dielectric substrate situated between the plurality of RFIC chips and attached to the second surface of the antenna substrate.

20. The antenna apparatus of claim **19**, wherein the antenna substrate comprises:

a first layer adjacent to the antenna elements;

a second layer proximate to the RFICs; and

an antenna ground plane between the first and second layers;

wherein the second layer comprises a transmission line coupling the plurality of RFIC chips to the BFN.

21. The antenna apparatus of claim **20**, wherein the antenna ground plane includes openings for first vias traversing therethrough, the first vias connecting the RFIC chips to the antenna elements; and

the second layer includes a plurality of second vias that couple the antenna ground plane to the RFIC chips.

22. The antenna apparatus of claim **1**, wherein the antenna substrate comprises a patterned metal layer that forms a beamforming network for the antenna apparatus.

23. The antenna apparatus of claim **1**, further comprising a plurality of serial peripheral interface (SPI) chips attached to the second surface of the antenna substrate and coplanarly arranged with and coupled to the RFIC chips, the SPI chips being coupled to the at least one circuit element through at least one of the solder columns.

24. The antenna apparatus of claim **1**, wherein the plurality of electrically conductive columns comprise the micro-coaxial cables.

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